1. What data type is the object below? 1 point

[1e-10,(1,2),"BGP",[3]]

- ○ Tuple
- ◉ List
- ○ Array
- ○ String

2. What is the value returned when the code below is executed: 1 point

```
>>> grades = [70,80.0,90,100]
>>> (grades[1]+grades[3])/2
```

- ○ 90
- ◉ 90.0
- ○ 80
- ○ 85.0

3. Suppose **splice_site_pairs** = ['GT-AG','GC-AG','AT-AC']. What is **splice_site_pairs**[:-1] ? 1 point

- ◉ ['GT-AG', 'GC-AG']
- ○ []
- ○ ['GT-AG','GC-AG','AT-AC']
- ○ ['GT-AG']

4. We want to add a new element to a list variable L. The new element is stored in a variable e. What command do we use? 1 point

- ◉ L.append(e)
- ○ L = L + e
- ○ L.addEnd(e)
- ○ L.insert(e)

5. Suppose t = ('a', 'c', 'g', 't'). What will be the output of the following code: 1 point

```
>>> t.append( ('A','C','G','T') )
>>> print len(t)
```

- ○ 8
- ◉ Error
- ○ 5
- ○ 2

6. What is the result of the print function in the following Python 3.xx code:

1 point

dna=input("Enter DNA sequence:")

dna_counts={'t':dna.count('t'),'c':dna.count('c'),'g':dna.count('g'),'a':dna.count('a')}

nt=sorted(dna_counts.keys())

print(nt[-1])

- error
- 't'
- 0
- 'a'

7. To delete an entry with the key 'a' from the dictionary dna_counts={'g': 13, 'c': 3, 't': 1, 'a': 16} what command do we use:

1 point

- dna_counts['a']=''
- del dna_counts['a']
- dna_counts[-1]=''
- dna_counts.delete('a':16)

8. Suppose dna is a string variable that contains only 'a','c','g' or 't' characters. What Python code below can we use to find the frequency (max_freq) of the most frequent character in string dna?

1 point

-
```
dna_counts=
{'t':dna.count('t'),'c':dna.count('c'),'g':dna.count('g'),'a':dna.count('a')}

max_freq=sorted(dna_counts.values())[-1]
```

-
```
dna_counts=
{'t':dna.count('t'),'c':dna.count('c'),'g':dna.count('g'),'a':dna.count('a')}

max_freq=dna_counts.sort()[-1]
```

-
```
dna_counts=
{'t':dna.count('t'),'c':dna.count('c'),'g':dna.count('g'),'a':dna.count('a')}

max_freq=sorted(dna_counts.keys())[-1]CODE
```

-
```
dna_counts=
{'t':dna.count('t'),'c':dna.count('c'),'g':dna.count('g'),'a':dna.count('a')}

max_freq=sorted(dna_counts.values())[0]
```

9. Suppose L1 and L2 are two list variables. What does the list variable

1 point

L3 = list(set(L1)&set(L2)) contain?

- All elements in lists L1 and L2
- A list of sets formed with the elements of lists L1 and L2
- All elements common to lists L1 and L2
- All elements common between lists L1 and L2 without duplicates

10. How many elements are in the dictionary someData after the following code has been executed?

1 point

```
someData = { }
someData['cheese'] = 'dairy'
someData['Cheese'] = 'dairy'
someData['Cheese'] = 'Dairy'
someData['cheese'] = 'Dairy'
```

- 2
- 3
- Can't say - an error message was issued.
- 0